

Ramya Yalamarthi

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 5+ years building production Generative AI (GenAI) systems, multi-agent architectures, and Retrieval-Augmented Generation (RAG) pipelines on AWS. Built Natural Language to SQL (NL-to-SQL) platforms cutting analyst query turnaround by 60%. Skilled in Large Language Model (LLM) orchestration, prompt engineering, MLOps, and cloud-native data engineering. M.S. in Information Technology.

EXPERIENCE

AI/ML Engineer

Nov 2025 – Present

QuinStreet, Seattle, USA

- Led development of a multi-agent NL-to-SQL system on AWS Bedrock AgentCore with Strands, orchestrating Claude Opus and Sonnet against Redshift - cutting analyst turnaround by 60%.
- Drove design of a RAG pipeline grounding SQL generation in schema, business glossary, and 50+ KPI definitions from DynamoDB and S3 - lifting accuracy 40% and eliminating hallucinations.
- Owned delivery of a self-healing NL-to-SQL engine on Redshift Serverless + Lambda with auto-retry, read-only SQL guardrails (DML/DDL + injection blocking), and pandas fallback - 35% accuracy gain.
- Directed the MLOps pipeline with AWS CDK, CodeBuild, Lambda, and ECR, automating ARM64 Docker images and Bedrock AgentCore deployments - cutting release turnaround from hours to 15 minutes.
- Diagnosed and patched a silent cost-misattribution bug in the Langfuse/OpenTelemetry model monitoring pipeline, recovering accurate cost tracking on ~30% of token volume.

Senior Product Engineer

Oct 2022 – Nov 2023

Simplilearn, Bangalore, India

- Prototyped Simplilearn's first Generative AI pilot using OpenAI GPT-3.5 and GPT-4 with Retrieval-Augmented Generation (RAG) over course content, enabling instant conversational help scoped to enrolled modules.
- Designed self-serve Tableau and Power BI dashboards on transactional and engagement data, tracking 20+ product KPIs across 5+ customer segments, eliminating 40% of ad-hoc analytics requests and reducing mid-course drop-off by 35%.
- Built ingestion pipelines on AWS integrating structured and unstructured data into Amazon Redshift, and standardized metric definitions across product, sales, and operations teams, cutting stakeholder reporting time by 30%.

Software Engineer

Jan 2022 – Sep 2022

Kristal.AI, Bangalore, India

- Built investor-facing features on a digital private wealth platform serving clients across Singapore, Hong Kong, India, and UAE using React, Node.js, and REST APIs, delivering a unified experience across four regulatory jurisdictions.
- Redesigned the advisor insights interface in React to present recommendations in a clear, actionable format for end clients, replacing cluttered legacy views with structured, easy-to-follow content.
- Analyzed user engagement with Python, SQL, and PySpark, finding under 5% open rate on time-sensitive market alerts during peak trading hours - reshaping alert timing and delivery decisions.

Technical Program Coordinator - Founding Team

Oct 2020 – Dec 2021

PeopleTech Group, Hyderabad, India

- Served as the primary bridge between the CEO and engineering teams during the launch of a new technology practice, translating executive priorities into actionable engineering plans and surfacing blockers, risks, and status to leadership.
- Partnered with engineering leads on scoping and reviewing client engagement plans, contributing to early technical decisions on AWS-based client data migrations and reporting workflows.
- Drove cross-functional alignment across engineering, client delivery, and business stakeholders, owning status and reporting cadence for the practice's client engagements as the team scaled.

Application Development Associate

Jul 2019 – Sep 2020

Accenture, Hyderabad, India

- Optimized Ab Initio data-processing pipelines and SAP ABAP interfaces across SAP ECC and CRM systems, using Oracle SQL for complex transformations, reducing batch processing time by 30%.
- Refactored legacy pipelines using CI/CD with GitHub Actions and automated testing, and implemented data quality controls (null checks, uniqueness validation, drift detection), reducing incident volume by 40% and improving pipeline reliability.

EDUCATION

M.S., Information Technology - Florida Atlantic University, FL, USA	Jan 2024 – Aug 2025
B.Tech, Computer Science - JNTU, Hyderabad, India	Aug 2015 – Jun 2019

CERTIFICATIONS

AWS Certified Developer - Associate (Mar 2026) | AWS Certified Generative AI Developer - Professional (Apr 2026)

TECHNICAL SKILLS

Programming & Frameworks: Python (FastAPI), SQL, PySpark, PyTorch, Hugging Face Transformers, LangChain, Strands Agents, NumPy, Pandas, React, Node.js

GenAI & LLMs: Generative AI, Large Language Models (LLMs), Multi-Agent Systems, AWS Bedrock, Bedrock AgentCore, Retrieval-Augmented Generation (RAG), Natural Language to SQL (NL-to-SQL), Prompt Engineering, Extended Thinking, Semantic Search, Vector Embeddings, OpenAI Embeddings, Claude Opus / Sonnet / Haiku, OpenAI GPT-3.5 / GPT-4, NLP (Named Entity Recognition, Summarization, Sentiment Analysis)

Cloud Platforms: AWS (Bedrock, S3, EC2, Lambda, Redshift, Redshift Serverless, DynamoDB, ECR, CloudWatch, CodeBuild, CDK), Azure (Databricks, Data Factory, Synapse, Event Hubs, Azure ML), Snowflake

MLOps & DevOps: AWS Cloud Development Kit (CDK), AWS CodeBuild, AWS Lambda, Amazon ECR, Docker, Terraform, GitHub Actions, CI/CD Pipelines, Experiment Tracking, Model Registry, Git

Data Engineering & Streaming: Apache Spark, Apache Spark Streaming, Apache Kafka, Event-Driven Architectures, ETL / ELT Pipelines, Real-Time Data Ingestion, Server-Sent Events (SSE), Redshift Data API, REST APIs

Observability & Responsible AI: Langfuse, AWS CloudWatch, Distributed Tracing, Token / Cost / Latency Metrics, Session Correlation, Model Monitoring, Model Evaluation, A/B Testing

Visualization & Reporting: Recharts, Tableau, Power BI, Jupyter Notebooks, Matplotlib, Seaborn, Plotly

Methodologies: Agile, Scrum, Software Development Life Cycle (SDLC)